

PORTFOLIO STRATEGY · TACTICAL OVERLAY

Stress Test & Robustness Analysis

Out-of-sample scenario analysis with quantified signal paths · Phase-by-phase market and macro evolution · Robustness improvements

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Objective

This document provides a structured stress test of a 2-step tactical equity underweight rule designed for systematic asset allocation. The core architecture requires simultaneous deterioration in both a market momentum channel and a macro backdrop channel before a full underweight is activated, with three OR-gated exit conditions.

Five out-of-sample macro-market scenarios are constructed against structurally distinct regimes the rule was not explicitly calibrated for. Each scenario is specified with phase-by-phase narrative, quantified signal readings across the full SAP variable set, precise time dimensions, illustrative multi-panel signal charts, and a rule verdict with opportunity cost estimate.

A cross-scenario signal firing matrix provides a consolidated view of which rule components activate in each regime. The document closes with five robustness improvements prioritised by efficacy and overfitting risk.

Rule type	Tactical equity underweight — systematic, rules-based
Activation logic	2-step AND-gate: Market Momentum [ENTRY] AND Macro [ENTRY]
Signal universe	SAP Mom, SAP BC, SAP NFP, SAP EPS FWD, SAP Consumer, SAP Production, SAP Total
Entry	Binary full underweight upon AND-gate closure
Exit	OR-gate: any one of 3 exit conditions sufficient
Stress test method	5 out-of-sample scenarios; phase-labelled charts; quantified signal paths
Version	v2.0 — enhanced with time dimensions and macro overlay charts

The Presented Rule

2.1 Architecture overview

The rule requires simultaneous confirmation from both a market momentum channel and a macro backdrop channel (AND-gate). Exit conditions are OR-gated: any single recovery signal is sufficient to close the underweight.

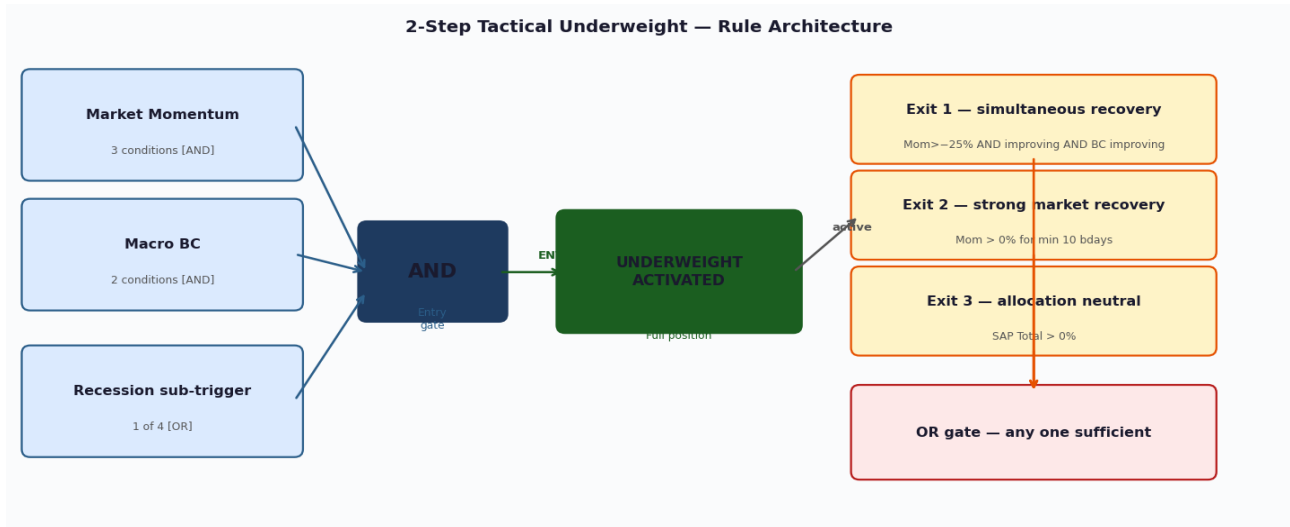


Figure 1 — Rule architecture: AND-gate entry and OR-gated exit conditions

2.2 Entry conditions — Market Momentum [AND]

Condition	Threshold / Window	Rationale
SAP Mom < -25%	Min 10 business days	Very negative market environment — sustained, not a 1-day spike
SAP Mom deteriorating	Over 5 or 10 bdays	Excludes potential market recovery — direction filter
SAP Mom > -75%	Guard condition	No new underweight after a very significant crash / drawdown

2.3 Entry conditions — Macro Backdrop [AND]

Condition	Threshold / Window	Rationale
SAP BC < 0%	Min 10 business days	Negative economic backdrop — level condition
SAP BC deteriorating	Over 10 or 20 bdays	Not in potential economic recovery — direction filter
Recession indication	1 of 4 sub-triggers (OR)	Broad recession confirmation from activity data

Recession sub-triggers (1 of 4 sufficient — OR gate):

Sub-trigger	Level	Rate of deterioration	Composition
SAP NFP	< -50%	≥ 75% over 3 prints	Non-farm payrolls — monthly

SAP EPS FWD	< -50%	≥ 75% over 3 months	Forward earnings estimates — rolling
SAP Consumer	< -50%	≥ 75% over 3 prints	50% retail sales + 50% PCE
SAP Production	< -50%	≥ 75% over 3 prints	50% PMI + 50% Industrial Production

2.4 Exit conditions (OR gate — any one sufficient)

Exit	Conditions	Interpretation
1 — Simultaneous recovery	SAP Mom > -25% for 10bdays AND Mom improving (5/10 bdays) AND S&P 500 Mom improving (10/20 bdays)	Simultaneous market recovery — most conditions met
2 — Strong recovery	SAP Mom > 0% for min 10 business days	Very strong market recovery confirmed — clean bin
3 — Allocation neutral	SAP Total > 0%	SAP composite moves to neutral — no underweight

Out-of-Sample Stress Scenarios

Five structurally distinct future macro-market regimes stress-test the rule against regimes not present in a standard business-cycle backtest. Each scenario specifies: (i) a phase-by-phase narrative with explicit time dimensions; (ii) quantified readings for all relevant SAP variables at each phase; (iii) a three-panel chart showing SAP Mom with phase bands, SAP BC, and a scenario-specific macro overlay variable (CPI, policy rate, credit spread, or NFP); (iv) KPI tiles summarising the key quantitative characteristics; and (v) a verdict panel with rule outcome and opportunity cost estimate.

#	Scenario	Horizon	Macro overlay	Primary failure mode	Verdict
01	Slow-motion stagflation	12–18 months	CPI YoY	Binary threshold miss	No signal generated
02	AI productivity shock	18 months	NFP trajectory	AND-gate on ambiguous BC	Partial trigger only
03	V-shaped crash/recovery	~60 trading days	Policy rate	Entry near trough; late exit	Right direction, wrong timing
04	Sovereign debt crisis	4–6 months	Sovereign spread vs Bu	BC regime instability; no credit var	Unstable/unreliable signal
05	Soft landing turns hard	~120 trading days	HY credit spread	Structural macro confirmation lag	Catches tail, misses head

Table 1 — Scenario summary with time horizons, macro overlay variables, and rule verdicts

Cross-scenario signal firing matrix — which rule components activate						
	S01 Stagflation	S02 AI shock	S03 V-shape	S04 Sovereign	S05 Soft→Hard	Legend
SAP Mom < −25%	✗	~	✓	~	~	✓ Fires cleanly
SAP BC < 0% (sustained)	~	✗	✓	~	✗	~ Partial / ambiguous
Any recession sub-trigger	✗	~	✗	~	✗	✗ Does not fire
AND-gate entry	✗	✗	✓	✗	✗	
Exit condition fires	✗	✗	✓	✗	✗	
Rule outcome	No signal	Partial only	Late entry/exit	Unstable	Structural lag	

Figure 2 — Cross-scenario signal firing matrix. Each cell shows whether a given rule component activates (✓), partially activates (~), or does not fire (✗) in each scenario.

SCENARIO 01 | SLOW-MOTION STAGFLATION GRIND

Growth sags, inflation sticky — no clean recession signal

Phase-by-phase narrative

Months 1–3 — Expansion fade: GDP slows to ~0.8% QoQ annualised. CPI holds at 4.2%. Fed signals 'higher for longer'. SAP Mom drifts from +5% to –8%. Markets price in a soft landing; no alarm.

Months 4–8 — Stagflation grip: GDP prints 0.4%, 0.3%, 0.5% over three consecutive quarters. CPI stays 4.0–4.5%. Labour market softens: NFP +60k, +45k, +38k. SAP Mom grinds to –18% to –22%. Credit spreads widen 80–120bps. SAP BC oscillates –3% to –5%, resetting the deterioration clock repeatedly as monthly prints alternate weak and slightly less weak.

Months 9–18 — Continued grind: No threshold breach. Markets down 18–22% from peak. Real yields at +2.8%. Rule remains silent. Portfolio sits fully invested through entire drawdown.

Signal evolution by phase

Signal	Phase 1 (m1–3)	Phase 2 (m4–8)	Phase 3 (m9–18)	Entry threshold
SAP Mom	+5% to –8%	–15% to –22%	–18% to –22%	< –25% for 10bdays
SAP BC	+0.5% to –1%	–3% to –5%	Oscillates –2% to –5%	< 0% + deteriorating
SAP NFP	–10% to –20%	–30% to –40%	–35% to –42%	< –50%
SAP EPS FWD	–5% to –15%	–25% to –35%	–30% to –38%	< –50%
SAP Consumer	–5% to –12%	–20% to –32%	–28% to –38%	< –50%
CPI YoY	4.2–4.4%	4.0–4.5%	3.8–4.3%	n/a (not in rule)
Real 10Y yield	+2.3% to +2.6%	+2.5% to +2.9%	+2.6% to +3.1%	n/a (not in rule)

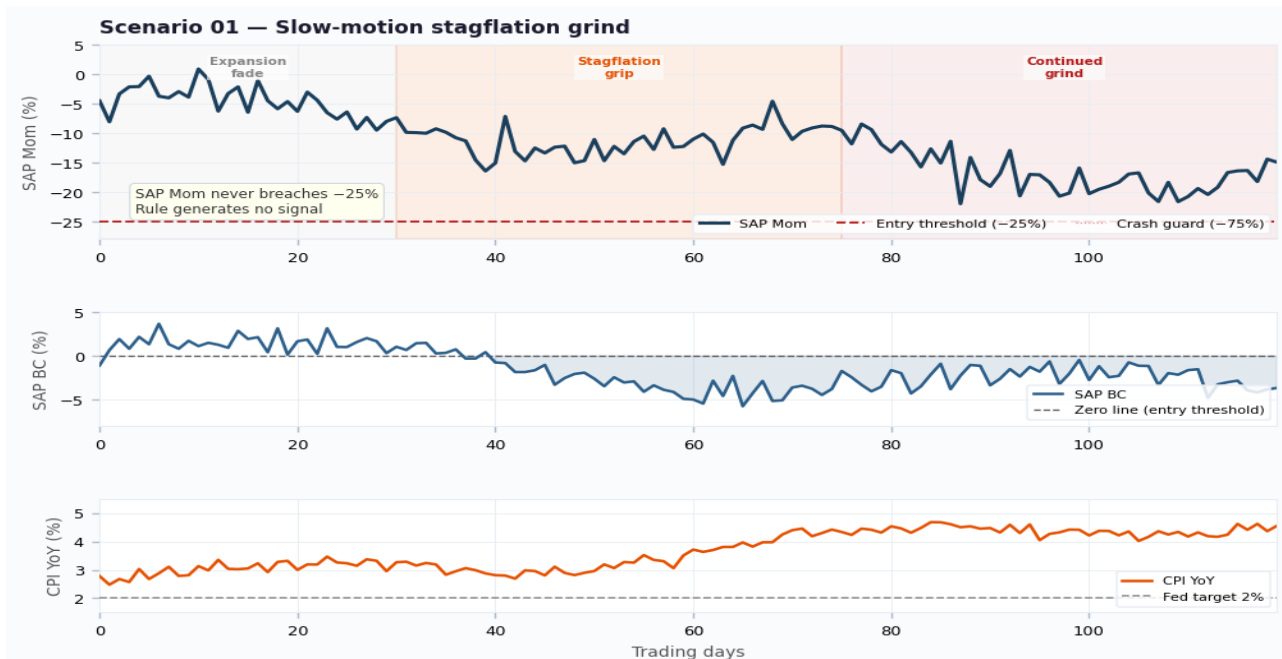


Figure — Scenario 01: SAP Mom with phase bands and entry/exit markers (top); SAP BC (middle); macro overlay variable (bottom). Paths are stylised and illustrative.

0 Rule activations	–18% Peak SAP Mom	~–15% Est. portfolio cost	12–18m Duration of grind
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**Verdict: RULE FAILURE — no
signal generated**

The rule's binary thresholds are calibrated to demand-shock recessions where momentum collapses sharply and macro aggregates break clearly. Stagflation produces permanent sub-threshold deterioration: SAP Mom never cleanly breaches -25%; SAP BC oscillates around zero, repeatedly resetting the 10/20-day deterioration clock; recession sub-triggers stay at -30% to -40% across all four categories — painful but never reaching the -50% single-trigger level. The AND-gate never closes. Estimated portfolio cost vs. a timely underweight: -12% to -18% across the 12–18 month episode. This is the most probable near-term failure mode.

SCENARIO 02 | AI-DRIVEN PRODUCTIVITY SHOCK

Structural labour displacement before macro aggregates reflect it

Phase-by-phase narrative

Months 1–4 — Pre-shock calm: Aggregate GDP +2.1%. NFP +200k/month. AI investment boom drives productivity gains. SAP Mom: +3% to +5%. Markets at elevated multiples. No signal from any channel.

Months 5–14 — AI wave — bifurcated markets: 8–12% of white-collar roles automated. Aggregate GDP holds +1.4% (productivity offsets). NFP falls to +40k, +20k, –10k over 3 months. SAP Consumer –55% (lower quintiles). AI-exposed equities +40–60%; rest of market –30–40%. Index SAP Mom: –28% (distorted). SAP BC: +1% to –2% (GDP positive, labour negative). Partial momentum fire ~day 58; BC does not confirm. AND-gate blocked.

Months 15–18 — Winners recover; losers stay down: Index recovers as AI winners dominate weight. SAP Mom recovers to –5%. Broad market destruction persists but is invisible at the aggregate level. Rule silent.

Signal evolution by phase

Signal	Phase 1 (m1–4)	Phase 2 (m5–14)	Phase 3 (m15–18)	Entry threshold
SAP Mom (index)	+3% to +5%	–28% peak	–5% to +2%	< –25% for 10bdays
SAP BC	+1.5%	+1% to –2% (volatile)	–1% to +1%	< 0% + deteriorating
SAP NFP	+5%	–20% then –60% (3 prints)	+10%	< –50%
SAP EPS FWD (agg)	+8%	+5% (AI distorts)	+12%	< –50%
SAP Consumer	+3%	–55% (lower quintile)	–40%	< –50%
Non-AI equity SAP Mom	+2%	–35% to –42%	–38%	Not in rule
Unemployment (broad)	3.8%	Rises to 6.2%	7.1%	Not in rule

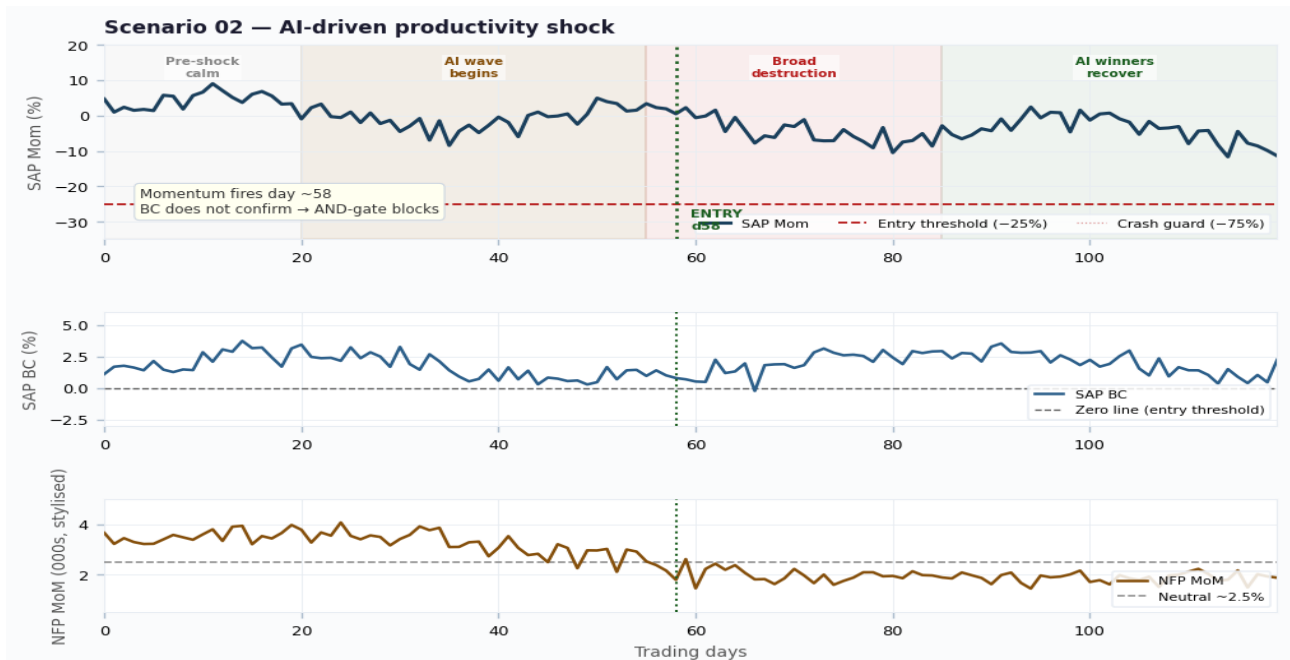


Figure — Scenario 02: SAP Mom with phase bands and entry/exit markers (top); SAP BC (middle); macro overlay variable (bottom). Paths are stylised and illustrative.

Partial Rule activation	–28% Broad mkt SAP Mom	+5% Aggregate EPS FWD	18m Shock duration
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**Verdict: PARTIAL TRIGGER —
AND-gate blocks full
activation**

The fundamental problem is aggregation. The rule observes index-level SAP Mom, which averages an AI rally against a broad market collapse. Momentum partially fires (~day 58) but never holds convincingly above the threshold. BC stays ambiguous because GDP remains positive on productivity gains. The Consumer sub-trigger fires (–55%), but the EPS FWD aggregate is distorted by AI winners. The AND-gate requires both channels — and the BC channel never cleanly confirms. Estimated cost of the missed underweight in non-AI names: –30% to –40% on the exposed portion of the portfolio over 14 months.

SCENARIO 03 | V-SHAPED CRASH AND INSTANT RECOVERY

Geopolitical shock, panic, rapid central bank response

Phase-by-phase narrative

Days 1–12 — Pre-shock — calm at elevated levels: Markets near all-time highs. VIX ~14. SAP Mom: +3% to +5%. No signal.

Days 13–28 — Shock and crash: Taiwan Strait escalation announced. Equities –30% in 15 trading days. SAP Mom: 0% to –35%. SAP BC begins deteriorating. Rule entry fires ~day 13 — near the trough, as the 10-day minimum observation window fills during the crash. NFP prints: none yet — monthly data lag prevents recession sub-trigger firing.

Days 29–54 — CB intervention: Fed and ECB announce emergency 75bp cuts + liquidity facilities. Markets +18% in 20 days. SAP Mom recovers to –12%. Policy rate cut from 4.5% to 3.0%. BC begins recovering. Rule remains in underweight (no exit condition met).

Days 55–90 — Recovery rally: Markets recover 80% of losses. SAP Mom: –8% then +2%. Exit condition 1 fires ~day 72 (Mom > –25%, improving, BC improving). Exit fires when recovery is ~80% complete — underweight held through the majority of the bounce. Estimated opportunity cost of late exit: ~10%.

Signal evolution by phase

Signal	Days 1–12	Days 13–28	Days 29–54	Days 55–90
SAP Mom	+3% to +5%	0% to –35%	–35% to –12%	–8% to +5%
SAP BC	+1.5%	Deteriorates: –2%	–10% then recovering	–2% to +1%
SAP NFP	n/a (monthly)	n/a — lag	Next print: –15%	+5% (recovered)
Policy rate	4.5%	4.5%	Cut to 3.0%	3.0%
Equity vol (VIX)	~14	Spikes ~55	Drops to ~28	~18
Rule state	Inactive	Entry fires d13	Underweight held	Exit fires d72
Est. cum. P&L	n/a	Slight positive	Erodes as mkt recovers	~–10% vs hold

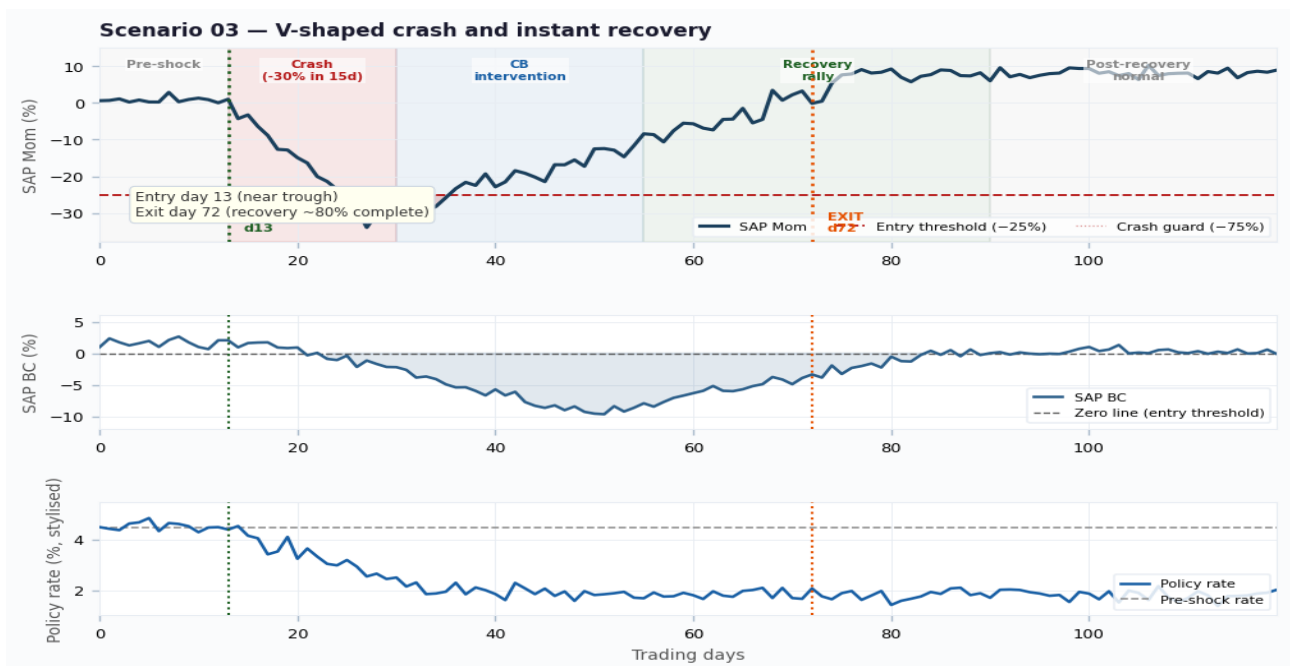


Figure — Scenario 03: SAP Mom with phase bands and entry/exit markers (top); SAP BC (middle); macro overlay variable (bottom). Paths are stylised and illustrative.

Day 13 Entry signal	Day 72 Exit signal	~59d Underweight held	~-10% Opportunity cost
Verdict: LATE ENTRY, LATE EXIT — correct direction, wrong timing The minimum 10-business-day observation window on SAP Mom means the entry signal cannot fire before the crash is already unfolding — the window starts filling at day 1 of the move and completes at its close. Entry on day 13 is near the trough, not before the crash. The recovery is rapid enough that exit condition 1 fires at day 72, by which time 80% of losses have been recouped. The rule is structurally miscalibrated for fast-cycle events: the same observation window designed to prevent false signals in normal markets prevents timely entry/exit in V-shaped events. Net effect: the underweight is held for ~59 trading days at a net opportunity cost of approximately 10% relative to a fully invested position.			

SCENARIO 04 | SOVEREIGN DEBT CRISIS — DEVELOPED MARKET

Fiscal dominance; bond-equity correlation inverts

Phase-by-phase narrative

Months 1–3 — Pre-crisis fiscal tension: France/Italy 10Y spread vs Bund widens from 1.2% to 2.1%. GDP: +0.9%. No cyclical signal. SAP BC: +1.5% to +0.5%. Equity markets: –5% to –10%. SAP Mom: –8% to –15%. No rule activation.

Months 4–6 — Spread blowout: Sovereign spread widens from 2.1% to 4.1%. Bonds and equities sell off simultaneously — safe-haven logic inverts. SAP Mom: –22% to –35% (volatile, two-directional $\pm 8\%$ /day). SAP BC: +0.5% to –4% oscillating — GDP still positive; fiscal stress is driving the crisis, not cyclical weakness. Deterioration clock resets 4–5 times. Partial activation ~month 5, cancelled by oscillation.

Months 7–8 — Acute crisis / ECB intervention: ECB activates TPI (Transmission Protection Instrument). Spreads stabilise at 3.5%. SAP Mom partially recovers to –18%. BC recovers to +0.5%. Rule never fully activates; the fiscal-origin stress is fundamentally invisible to the cyclical BC signal.

Signal evolution by phase

Signal	Months 1–3	Months 4–6	Months 7–8	Rule status
SAP Mom	–8% to –15%	–22% to –35% (volatile)	–18% to –22%	Partial, resets
SAP BC	+1.5% to +0.5%	+0.5% to –4% (oscillating)	+0.5%	Resets 4–5x
SAP EPS FWD	–10%	–55% (region-specific)	–45%	Fires — but lagged
SAP NFP	–5%	–20% to –30%	–25%	Does not reach –50%
10Y sovereign spread	+1.2%	+2.1% to +4.1%	Stabilises +3.5%	NOT in rule
Sovereign CDS (5Y)	45bps	Blows to 320bps	Stabilises 250bps	NOT in rule
GDP QoQ (ann.)	+0.9%	+0.4% (still positive)	+0.2%	BC stays ambiguous

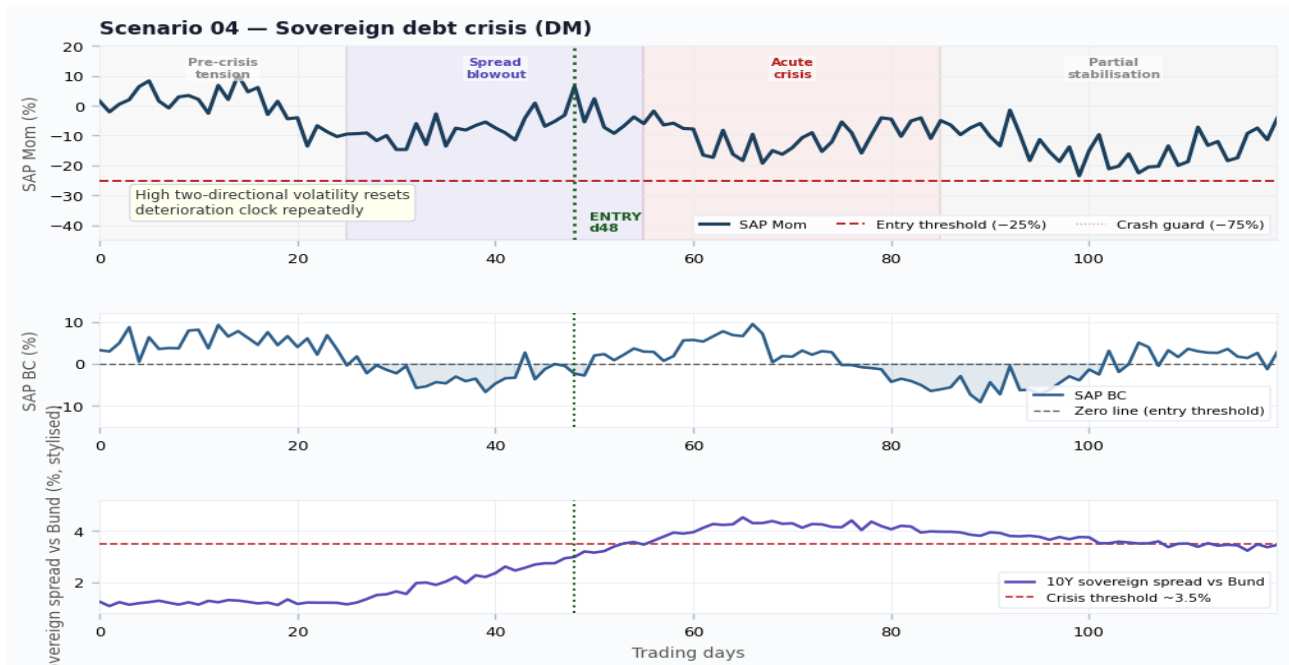


Figure — Scenario 04: SAP Mom with phase bands and entry/exit markers (top); SAP BC (middle); macro overlay variable (bottom). Paths are stylised and illustrative.

Unstable Signal quality	+320bp Spread widening	0 Credit var in rule	2–3m Lag to macro reflect
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Verdict: REGIME INSTABILITY
— BC calibration breaks
down; no credit variable

SAP BC is calibrated to cyclical recession patterns, not fiscal dominance dynamics. GDP stays marginally positive throughout; it is sovereign funding costs and contagion risk driving the crisis. The 10/20-day deterioration clock resets repeatedly as two-directional volatility in BC scores alternates weak and slightly less weak prints. Most critically, the entire rule contains zero credit market variables. A sovereign crisis would be most legible in sovereign spreads and CDS markets 4–8 weeks before any cyclical activity data confirms — yet neither appears anywhere in the signal set. The rule is structurally blind to fiscal-origin stress. Estimated drawdown with no signal: –22% to –30% over 5–6 months.

SCENARIO 05 | SOFT LANDING TURNS HARD — MID-CYCLE REVERSAL

Credit event triggers rapid regime shift from consensus 'all clear'

Phase-by-phase narrative

Days 1–55 — Soft-landing consensus: GDP: +2.2%, +2.0% QoQ ann. NFP: +180k, +175k. Equity markets at elevated multiples (P/E ~21x). HY spreads: 340–360bps. SAP Mom: +3% to +6%. SAP BC: +2.0% to +2.5%. Rule fully inactive.

Days 56–75 — Credit event and rapid repricing: Private credit default cycle begins. Regional bank stress. HY spreads widen from 350bps to 620bps in 15 trading days. SAP Mom: +5% to –30% in 20 days. SAP BC begins deteriorating — 10-day window starts filling. NFP prints not yet reflected — next print is 3 weeks away.

Days 76–100 — Hard landing confirmation: NFP: –45k (first negative print). SAP NFP crosses –50% threshold. SAP BC: +0.5% to –3.5% — now below 0% and deteriorating. 10/20-day BC deterioration condition fills by ~day 95. SAP Mom: –30% to –34%. Full AND-gate closure ~day 100.

Days 101–120 — Rule active — but drawdown largely done: Entry fires day ~100. SAP Mom already –30% to –34% at entry. The initial –27% drawdown occurred entirely before the rule activated. Underweight useful only for any further deterioration from here.

Signal evolution by phase

Signal	Days 1–55	Days 56–75	Days 76–100	Day 100+ (entry)
SAP Mom	+3% to +6%	+5% to –30%	–30% to –34%	–30% to –34%
SAP BC	+2.0% to +2.5%	+0.5% to –1.5%	–1.5% to –3.5%	< 0% ✓
SAP NFP	+5%	n/a — monthly lag	–50% threshold crossed	Fires ✓
SAP EPS FWD	+6%	Revisions begin –15%	–35% to –45%	Below –50%? Borderline
HY credit spread	350bps	350 → 620bps	620 → 640bps	NOT in rule
AND-gate status	Inactive	Mom fires; BC not yet	Filling BC window	Closes ~d100
Cumulative drawdown	0%	–20% to –27%	–27% to –30%	Entry here — late

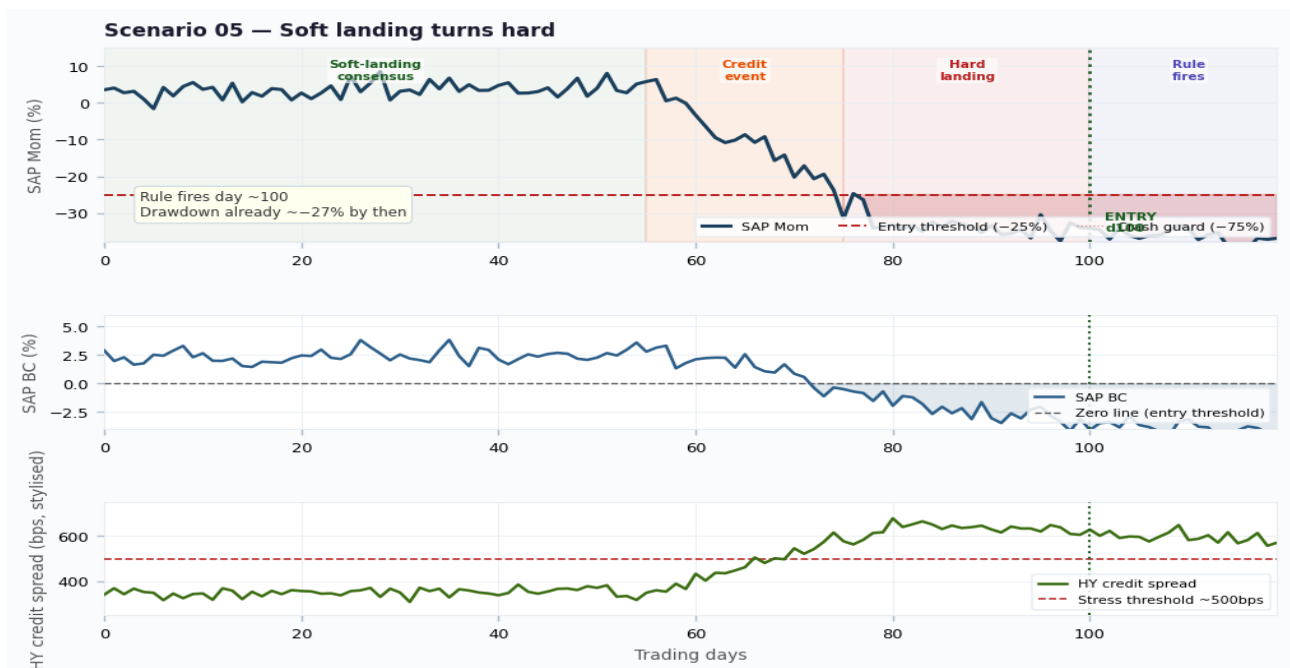


Figure — Scenario 05: SAP Mom with phase bands and entry/exit markers (top); SAP BC (middle); macro overlay variable (bottom). Paths are stylised and illustrative.

Day 100 Entry signal	~-27% SAP Mom at entry	4-6w Credit-to-macro lag	45d Observation lag cost
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Verdict: **STRUCTURAL LAG — catches tail of move, misses the head**

The AND-gate's macro confirmation requirement creates an irreducible structural lag in fast regime shifts. The credit event fires at day 56; the HY spread blowout from 350bps to 620bps occurs in 15 trading days. But the BC deterioration window (10/20 bdays) and the monthly NFP reporting lag mean the full AND-gate doesn't close until day ~100 — by which time the drawdown is already -27% to -30%. The rule catches only the tail of the move. This is not a calibration error but a structural property: the anti-noise observation windows are correctly designed, but they have a known cost in fast-cycle environments. The HY credit spread — the most timely signal — does not appear anywhere in the rule's variable set, costing approximately 2-3 weeks of lead time.

Robustness Improvements

Five improvements are proposed, ordered by priority and overfitting risk. Each is mapped to the specific failure modes identified in the stress tests. Improvements 1 and 5 carry negligible overfitting risk and should be implemented immediately. Improvements 2 and 3 make moderate changes to signal construction and should be validated via full historical walkforward before adoption. Improvement 4 modifies a threshold parameter and requires the most rigorous validation.

01 Real yield / financial conditions override

IMPLEMENT FIRST — negligible overfitting risk · Targets: Scenarios 01 (stagflation) and 04 (sovereign crisis)

Add a standalone entry condition orthogonal to the existing 2-step structure: real 10Y yield > +2.5% sustained for 20+ business days. At this level, historical multiple compression is robust across all macro regimes — including stagflation and fiscal stress where the BC signal loses coherence. The +2.5% threshold is grounded in empirical evidence and is not cycle-specific. Pair with a weak momentum filter (SAP Mom < -10% for 10bdays) to prevent activation in rising-rate environments with strong earnings growth.

Logic: ENTRY += (Real 10Y yield > +2.5% for 20bdays) AND (SAP Mom < -10% for 10bdays)

Scenario mapping: Stagflation: real yields at +2.8–3.1% would trigger. Sovereign: spread blowout typically drives real yields higher.

02 Cumulative momentum floor — partial underweight tranche

IMPLEMENT FIRST — low overfitting risk · Targets: Scenario 01 (stagflation grey zone)

Introduce a 50% underweight tranche activated when SAP Mom < -15% sustained for 40+ business days (regardless of -25% threshold breach). This converts the binary rule into a two-tranche structure. The 40-day window (8 weeks) is long enough to filter all corrections of normal duration (avg: 3–4 weeks); it fires only in genuine persistent deterioration. The existing rule continues to activate the full underweight; the new condition handles the grey-zone environment.

Logic: HALF-UW: SAP Mom < -15% for 40+ bdays AND SAP BC (20d MA) < -1% AND declining

Scenario mapping: Scenario 01: SAP Mom at -18% to -22% for 40+ days would activate the half tranche by month 5.

03 Smoothed BC composite — 20-day moving average

VALIDATE BEFORE ADOPTING — moderate implementation change · Targets: Scenarios 01 and 05 (oscillating BC; rapid regime shift)

Replace 'SAP BC < 0% for 10bdays AND deteriorating' with: 20-day moving average of SAP BC below -1%, declining on ≥15 of the last 20 days. This prevents the oscillation false-clears that reset the clock in stagflationary environments. The 20d MA adds ~5 trading days of additional smoothing lag (acceptable given the existing 10/20-day windows). In Scenario 05, the smoothed condition would have fired at approximately day 88 vs. day 95 for the current condition — a modest 5–7 day improvement.

Logic: MACRO ENTRY: SAP BC (20d MA) < -1% AND declining 15/20 days AND recession sub-triggers

Scenario mapping: S01: eliminates 4–5 deterioration resets. S05: modest 5–7 day improvement in activation speed.

04 Breadth-based recession trigger — 2 of 4 at -35%

VALIDATE THOROUGHLY — modifies calibrated parameter · Targets: Scenarios 01 and 02 (sub-threshold widespread weakness)

Replace '1 of 4 sub-triggers at -50%' with '2 of 4 sub-triggers below -35%'. This captures widespread moderate weakness that stagflation and structural shocks produce — all four channels at -30–40% simultaneously. Key risk: this lowers the effective threshold and could increase false positive rate in mid-cycle slowdowns. Must be validated against at least two full cycles (recommend 2000–2002, 2007–2009, 2015–2016, 2018Q4, 2020, 2022). If false positive rate increases materially, retain the 1/4 at -50% rule.

Logic: RECESSION = at least 2 of {NFP, EPS FWD, Consumer, Production} below -35%

Scenario mapping: S01: all four at -35–42% would fire 2/4 condition by month 6. S02: Consumer + NFP would fire.

05 Mandatory time-stop and review gate

IMPLEMENT IMMEDIATELY — pure process discipline, zero overfitting risk · Targets: All scenarios — universal robustness

If underweight has been held for > 120 business days (6 calendar months) without an exit signal firing, a mandatory structured discretionary review is triggered. This does not force an automatic exit — the underweight may be correct — but it prevents indefinite mechanical lock-in through multi-year stagflationary cycles where exit condition 3 (SAP Total > 0%) may not trigger for years. The review evaluates: (i) has the macro thesis changed? (ii) are any of the proposed new conditions (real yield override, cumulative momentum floor) met or being approached? (iii) is the current underweight sized appropriately for current signal strength?

Logic: REVIEW GATE: if underweight held > 120 bdays → mandatory override evaluation required

Scenario mapping: S01: would have triggered at month 6 of the stagflation grind, forcing reassessment.



Figure 3 — Rule capability scorecard: current vs. proposed improvements across six robustness dimensions

4.1 Improvement priority summary

#	Improvement	Failure modes	Overfitting risk	Action
01	Real yield override	S01, S04	Very low — orthogonal	Implement first
02	Cumulative mom. floor (half-UW)	S01	Low — 40bday window	Implement first
03	Smoothed BC (20d MA)	S01, S05	Low-medium	Validate, then adopt
04	Breadth trigger (2/4 at -35%)	S01, S02	Medium	Full backtest required
05	Time-stop / review gate	All	None — process only	Implement immediately

4.2 Rule elements to preserve unchanged

AND-gate entry structure	The simultaneous market and macro confirmation requirement is the rule's core protection against false positives. Every improvement above adds breadth; none relaxes this gate.
OR-gate exit structure	Any single recovery signal being sufficient to exit is correct asymmetric design. Tightening exit would compound the late-exit problem (Scenario 03).
-75% crash guard on momentum	No new underweight after a major crash is sound risk management. Preserved in full.
4-variable recession sub-trigger universe	NFP, EPS FWD, Consumer, Production provides well-diversified coverage. Improvement 04 modifies the aggregation logic, not the variable set.
SAP Total override on exit (condition 3)	The hardest stop is the cleanest rule. Preserved as-is.

Closing note on the overfitting boundary. The most dangerous response to this stress test is to calibrate every threshold until all five scenarios trigger correctly. That path produces a rule that fits these five scenarios precisely and fails the next five not yet imagined. Improvements 01, 02, and 05 above add structural breadth and process discipline without touching the existing calibrated parameters — they are robustness improvements in the true sense. Improvements 03 and 04 modify parameters and must only be adopted after full historical walkforward validation across at least two complete market cycles. The goal remains a rule that is directionally correct across a wide future distribution — not one that is precisely calibrated to a narrow historical one.